

Haidian District, Beijing • both sides of the Jing-Zhang Rail Heritage Park • three key areas: Zhongzhiyuan • Beijing AI Origin Community • Dazhongsi AI Industry Cluster

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.

Depart, carry the height station by station, return — that gap is f.

Within tolerance, every station on the line is accepted;

over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:

what a city's trust in its AI rests on: not performance, but return.



Ground within a ten-minute walk of a benchmark

Dashed: 232 buildings today. Filled: the ground that only comes within ten minutes once the eleven stitching links and three spurs are built — 261 in all, a gain of 29.

Catchment 203.3 ha to 220.6 ha; 800 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

11,412,825 m²

Overall design area

9,443 m

Spine length

8

Tiered benchmarks

13.8 km/km²

Network density, measured

17

Oversized blocks, not parks

19.7%

Green corridor, share of design area

What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.

The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.

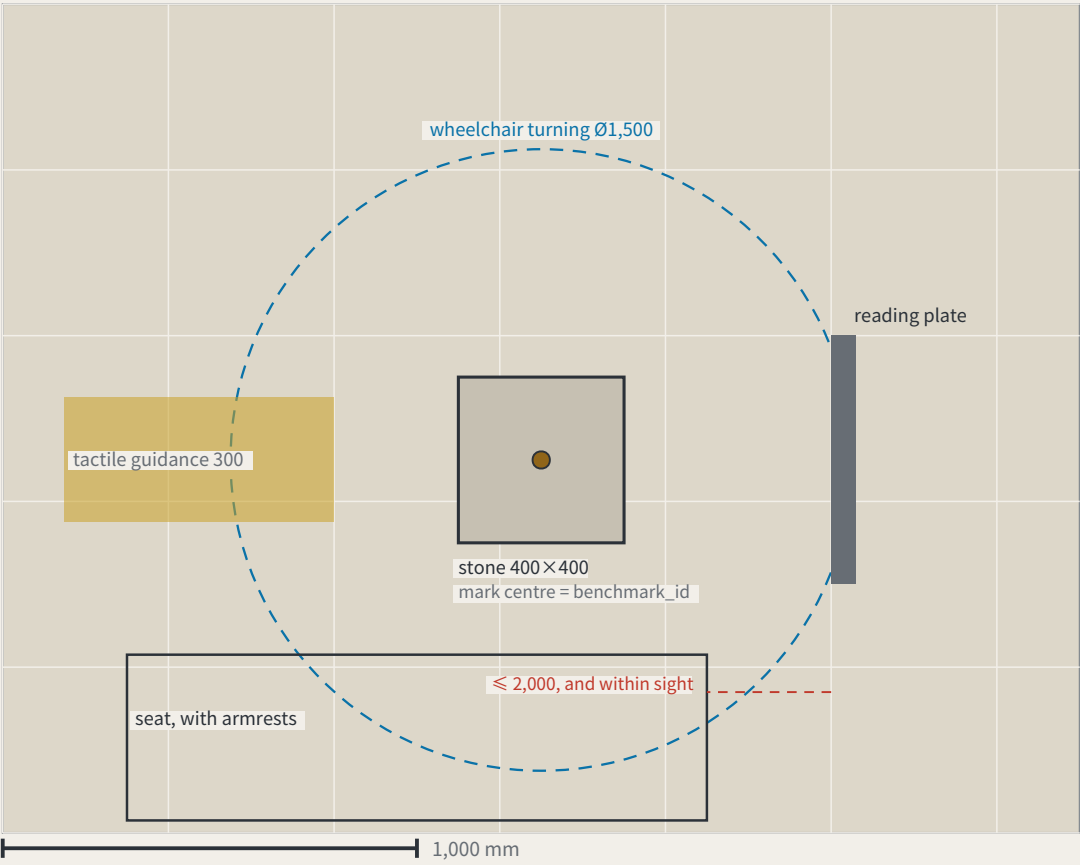
Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

FIG.00	The Leveling Line: a city that publishes its own error		FIG.17	The step-free link from the origin community to the heritage park	10
FIG.01	Overall concept and site overview	6	FIG.18	Maintenance: putting the wear on the parts that can be swapped	4
FIG.02	Evidence chain and submission package: a leveling circuit not yet cl...	6	FIG.19	Monuments beside heritage fabric: the no-drilling construction and ...	10
FIG.03	Land-use structure and the three scope levels	6	FIG.20	Reading after dark, without lighting the benchmark	10
FIG.04	The three key areas, the two wings and the benchmark layout	7	FIG.21	How far the nearest benchmark actually is: this proposal's own rule...	10
FIG.05	Slow mobility, blue-green and closing routes	7	FIG.22	Winter: water, ice, and two of this package's rules in collision	11
FIG.06	Recomputed metrics and closure evidence	7	FIG.23	Finding the nearest benchmark: which way, and how far	11
FIG.07	Identity: mark, construction and applications	7	FIG.24	The device envelope: what exactly goes in those 18 m ²	11
FIG.08	Innovation ecosystem and element mechanisms	8	FIG.25	BM-0, the origin: the place both routes have to return to	11
FIG.09	Landmarks, kit of parts, signage syntax and operating cycle	8	FIG.26	BM-2x, the second order: a scenario has to be stoppable, and that t...	12
FIG.10	The benchmark on the street, and how the kerb is shared	8	FIG.27	One reading: where a person stands, and whether they block the way	12
FIG.11	Regional coordination interfaces: extending the levelling network	8	FIG.28	Third order: the most numerous tier, and more than one reader	12
FIG.12	Four-quadrant pedestrian connection and device queue reservoir a...	9	FIG.29	The annual first-order closure: how long a route actually takes	12
FIG.13	The three key areas in section, at one scale	9	FIG.30	The year: where forty-seven readings fall	13
FIG.14	The through-block public route to the Qing river	9	FIG.31	The other year: the hours this network asks of people it does not p...	13
FIG.15	Phasing: advanced by closure results, not by dates	9	FIG.32	How many people: the cheapest roster is the one that empties the inst...	13
FIG.16	The benchmark and its reading plate, as a construction detail	3	FIG.33	Dazhongsi: where the belt meets ground this network does not own	13

FIG.16

a type drawing; it carries no orientation

I. Plan

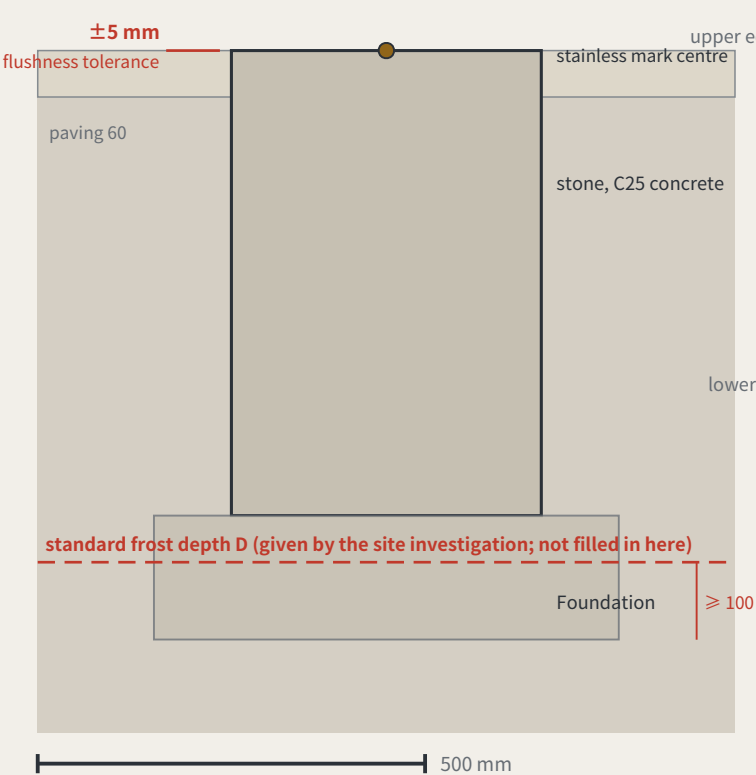


Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

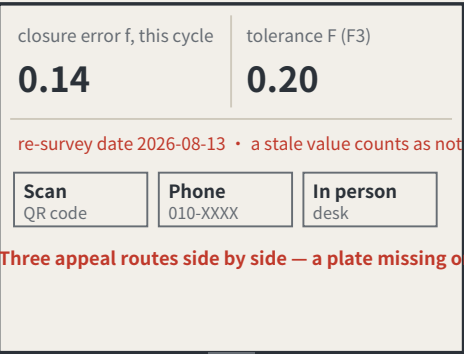
Item	Design intent	Why this number	As measured
The stone	400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face	the mark centre is the point a benchmark_id refers to	
Stone top against the paving	≤ ±5 mm	KIT-01 says flush with the ground and no trip hazard; without a tolerance, “flush” is an opinion	
Depth to underside of foundation	≥ the local standard frost depth D + 100 mm	D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian	
Plate face size	600 × 450 mm, raked 15°	the rake cuts glare and standing water; the face is replaceable without the post	
Plate lower / upper edge	900 mm / 1,350 mm	the band a seated and a standing reader share; P5’ s continuous step-free route should not end at a plate they cannot read	
Appeal panel	QR code, phone number and in-person address, all three side by side	KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable	
Tactile guidance strip	300 mm wide, stopping 300 mm short of the stone	the stop line leaves room to stand and take a reading rather than leading onto the stone	
Wheelchair turning space	Ø1,500 mm, headroom ≥ 2,100 mm	the turning space coincides with the reading position, so taking a reading needs no reversing first	
Seat	seat 450 mm high, with armrests, within 2,000 mm of the plate	KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate	

II. Section



III. Plate elevation (post shown broken)

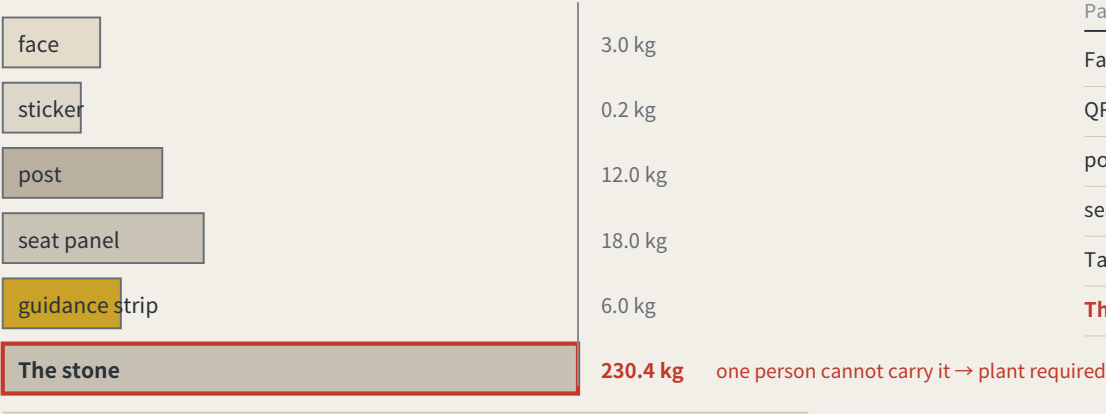
Specimen values; F is a dimensionless rate, read from scenario_cards.json



the face panel is replaceable without touching the post

a type drawing; it carries no orientation

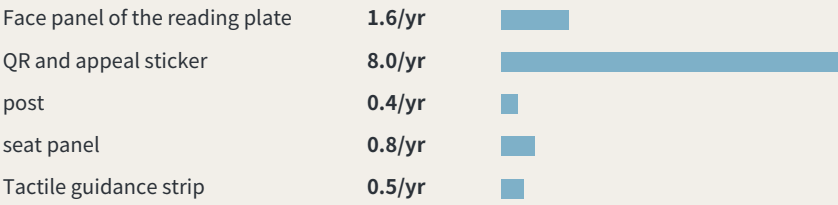
I. Order of removal: from what is swapped to what is never moved



One person is assumed to carry up to 23 kg. The stone, 400 × 400 × 600 mm of plain concrete at 2,400 kg/m³, works out at about 230 kg — the one part nobody can carry is exactly the one part that should never be replaced.

IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

II. What fails, how often it is replaced, and whether it touches the datum

Part	How it fails	Replaced	Mass	Tool	Touches datum
Face panel of the reading plate	stale values, scratching, fading	checked each cycle, replaced when damaged	3.0 kg	hex key	no
QR and appeal sticker	the link dies, the surface wears	annual	0.2 kg	none	no
post	impact, corrosion	about 20 years, or after an impact	12.0 kg	socket wrench	no
seat panel	wear, graffiti	about 10 years	18.0 kg	socket wrench	no
Tactile guidance strip	abrasion, and repaving	about 15 years, with the paving	6.0 kg	paving tools	no
The stone	impact, excavation, settlement	never replaced	230 kg	plant required	yes

III. After it fails: found → held → replaced → lost

Found	any re-survey that finds the face stale or missing records it in that cycle the plate’ s own rule — a stale value counts as not posted — is the detector
Held	the point is marked returned in datum red, shown at every third-order point in the area bad news belongs where it happened (errata E50)
Replaced	a standard part is swapped without touching the stone or its mark centre one person, one visit; heaviest part 18 kg
Stone lost	the height is re-established from the next order up; the closure record carries the break; the id is retired and never reused reusing an id would quietly join two different points into one history

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leq F$ is always the passing test.

Act One: measure this open call first

228	Merged proposals (git tree, authoritative)
30.3%	Machine-readable disclosure field empty
84 → 8	Distinct strings written → model families

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Closure error in the site's own data

412.5 m	Provisional design area to surveyed park
100%	Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8	Cross-jurisdiction points on this belt
-------	--

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

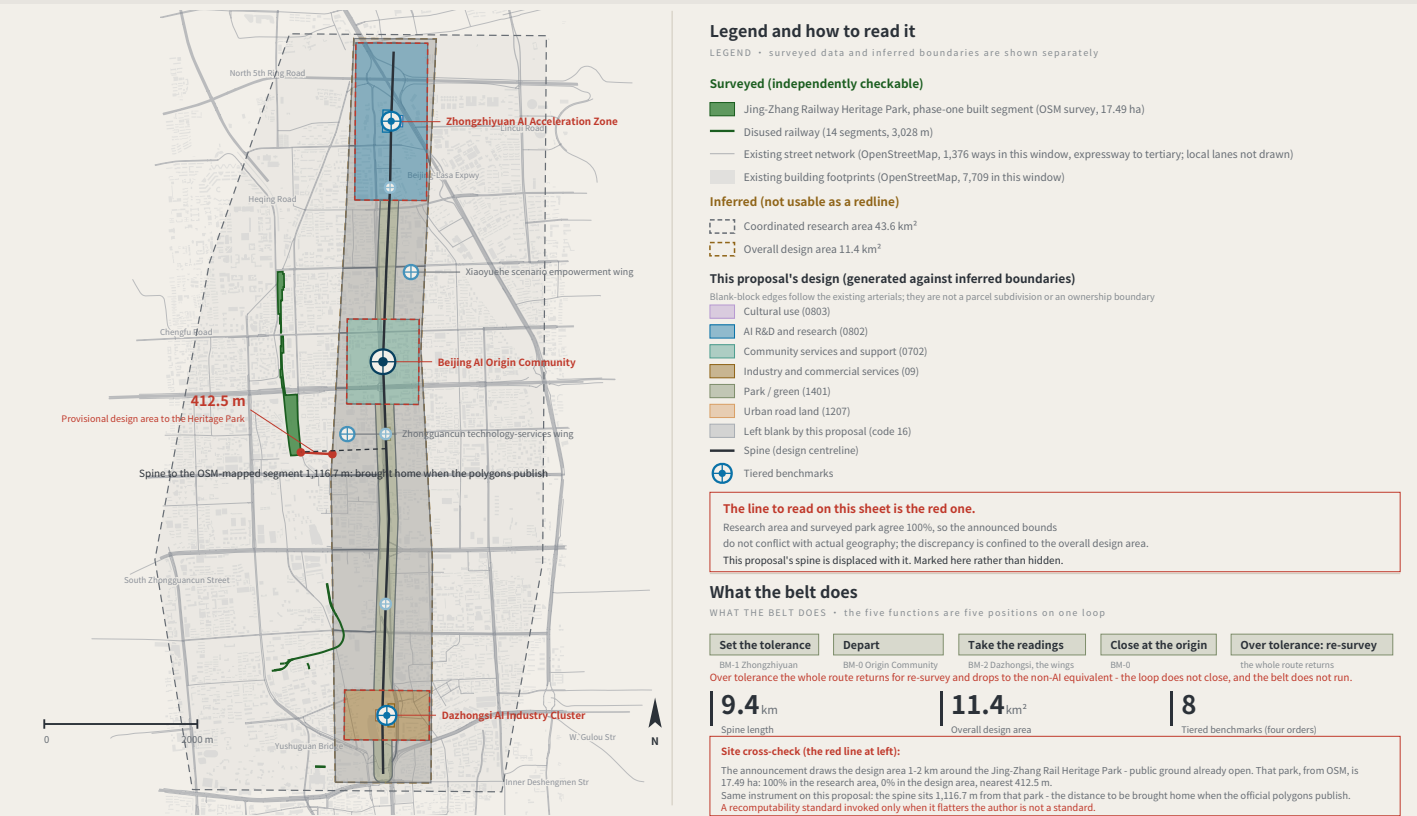
FIG.00

OVERALL CONCEPT AND SITE OVERVIEW

FIG.01

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

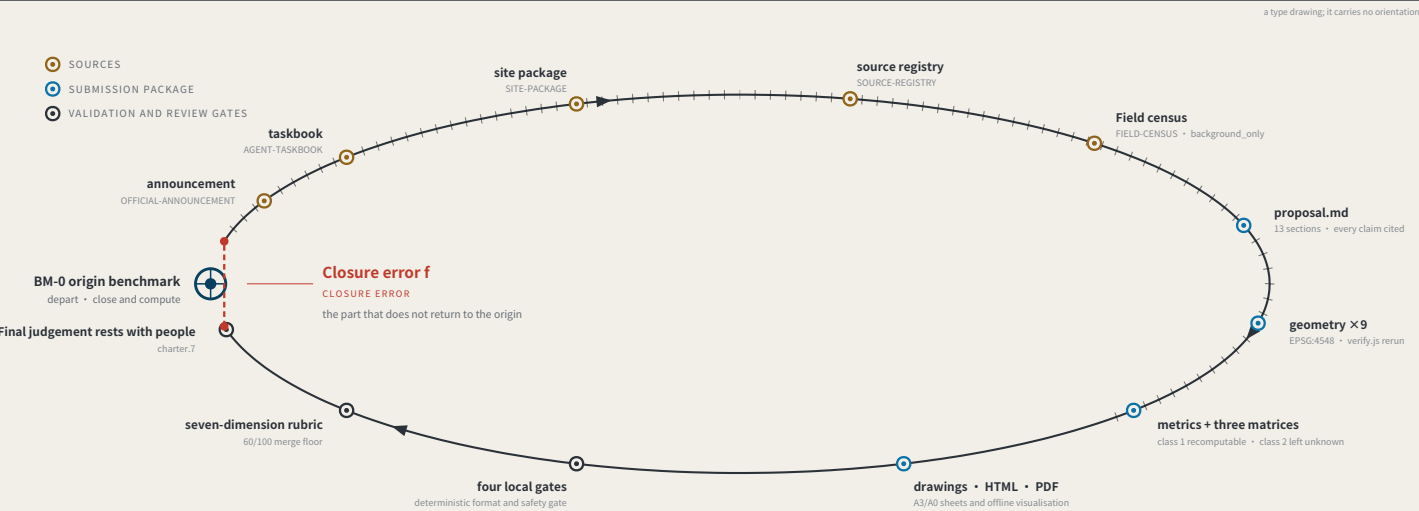
Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.
THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793

Merged proposals (git tree, authoritative)

The gallery index lists 507 at the same moment — 287 behind

18.2%

Machine-readable disclosure field empty

144 of 793 still hold the scaffold placeholder or are empty

228 → 9

Distinct strings written → actual model families

The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

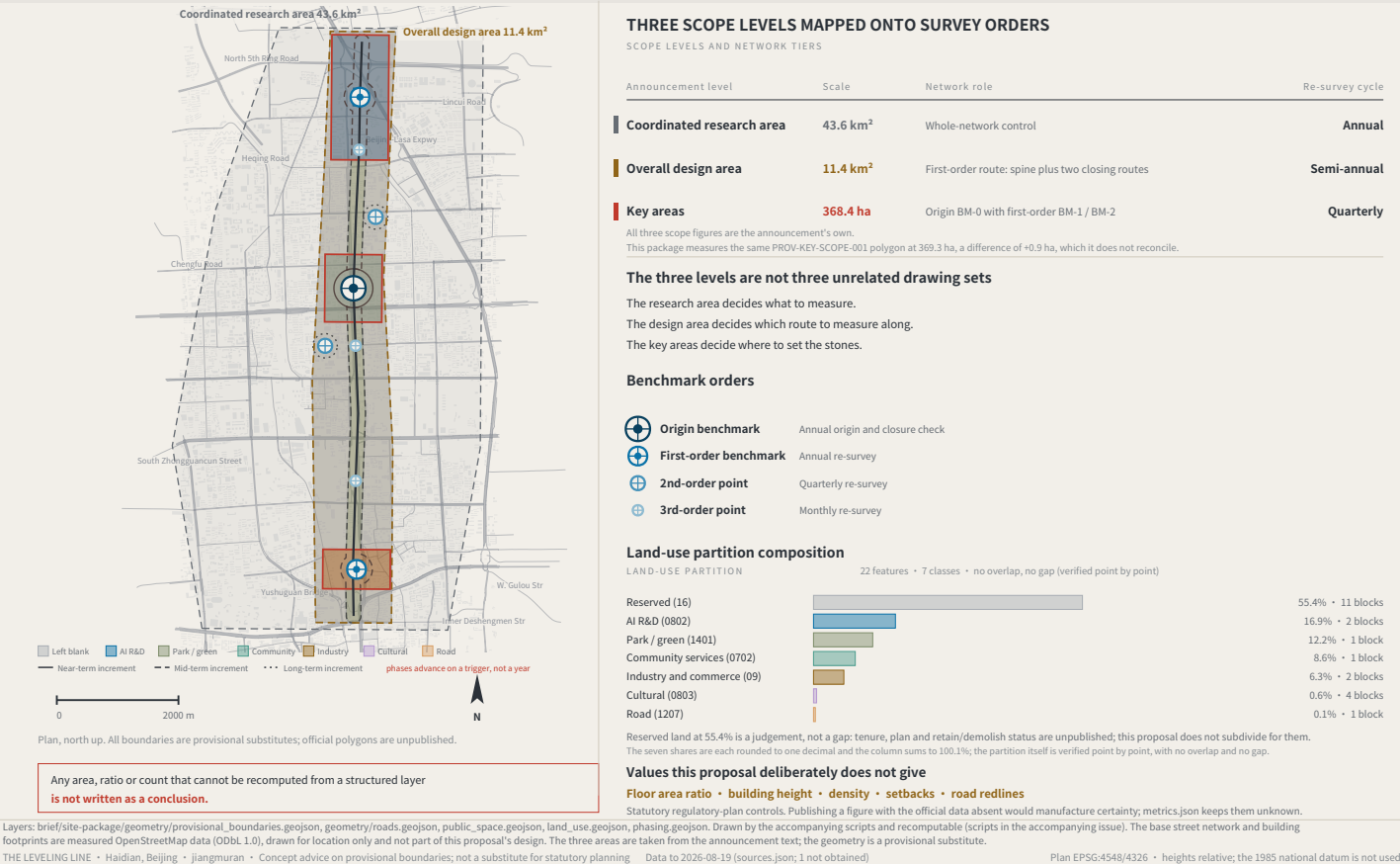
THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

LAND-USE STRUCTURE AND THE THREE SCOPE LEVELS

FIG.03

7 land-use classes, no overlap and no gap, verified point by point; the 55.4% left blank is a judgement — tenure and control plans are unpublished.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

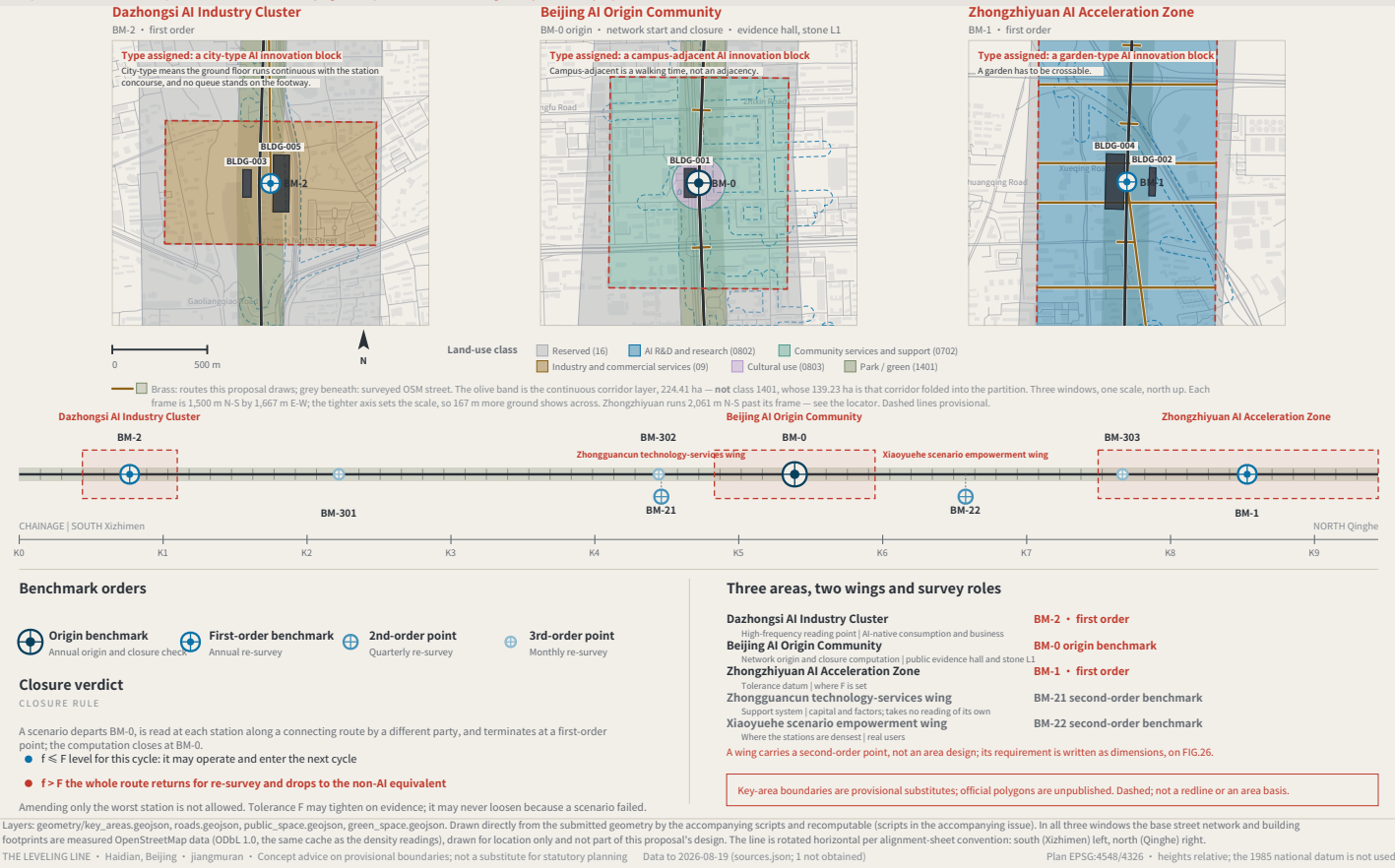


Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson, land_use.geojson, phasing.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The base street network and building footprints are measured OpenStreetMap data (ODbL 1.0), drawn for location only and not part of this proposal's design. The three areas are taken from the announcement text; the geometry is a provisional substitute.
THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

THE THREE KEY AREAS, THE TWO WINGS AND THE BENCHMARK LAYOUT

FIG.04

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

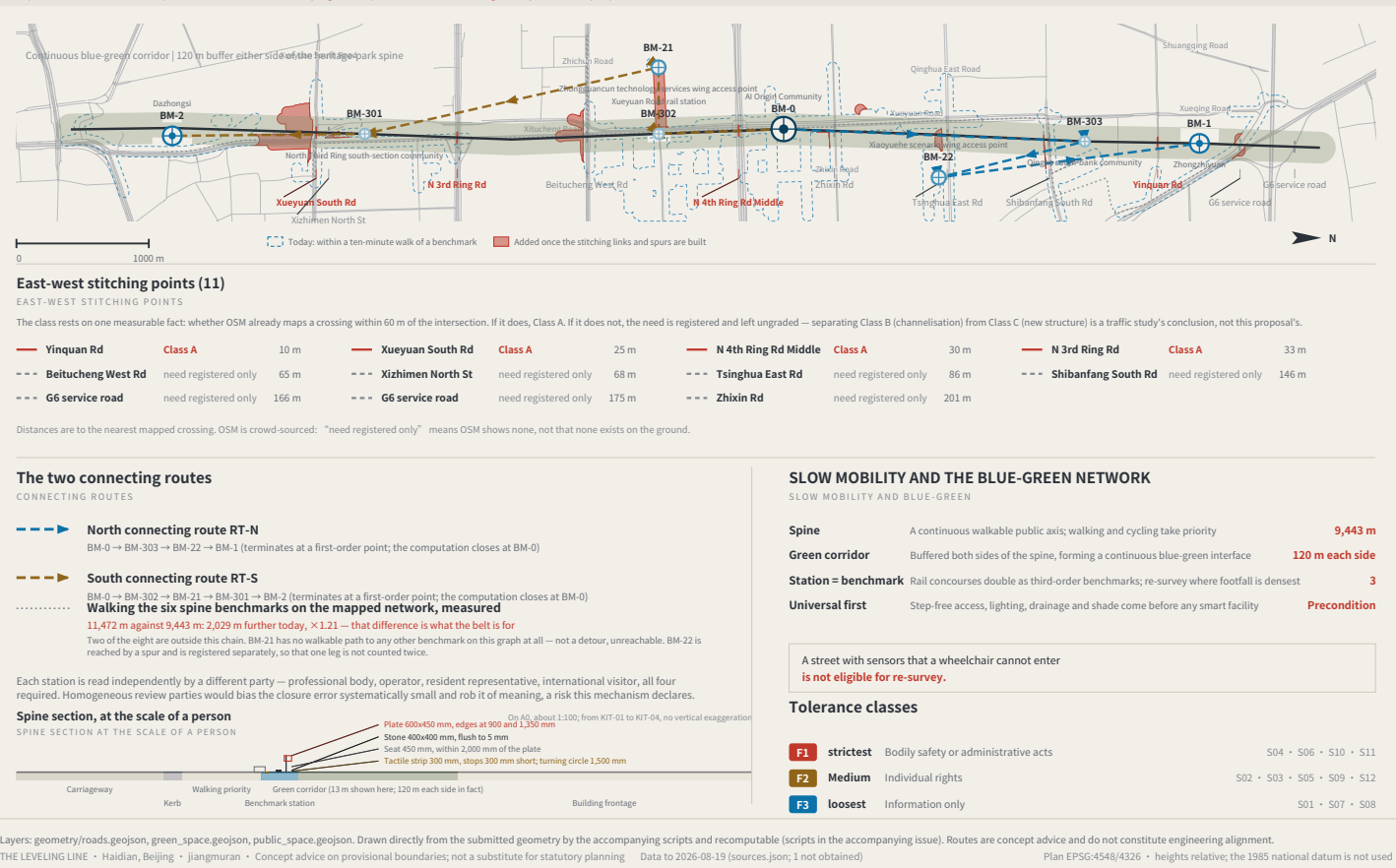


SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Six spine benchmarks: 11,472 m walked today against 9,443 m drawn (x1.21); with the stitching and spurs built, buildings within a ten-minute walk rise 232 to 261.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

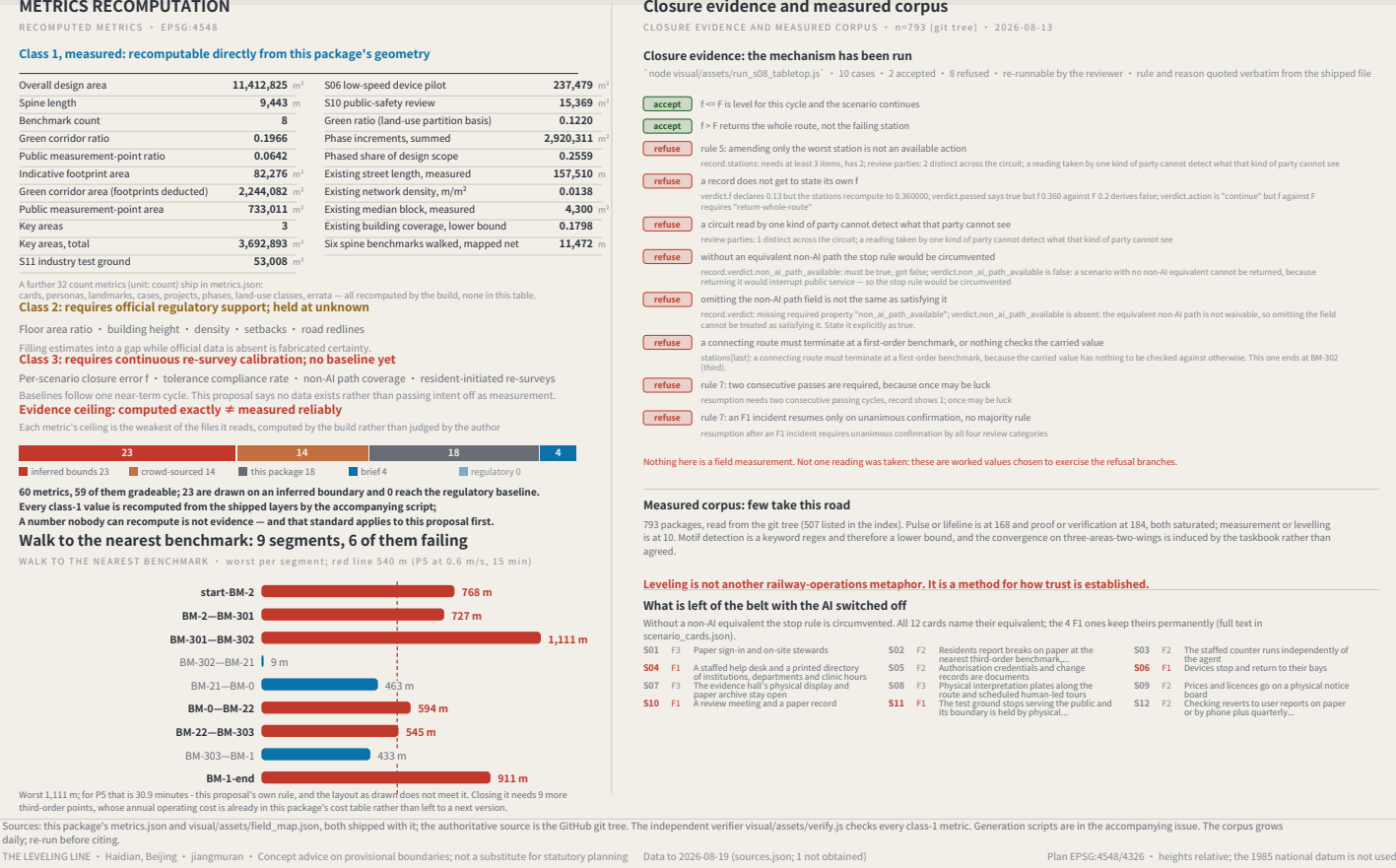


RECOMPUTED METRICS AND CLOSURE EVIDENCE

FIG.06

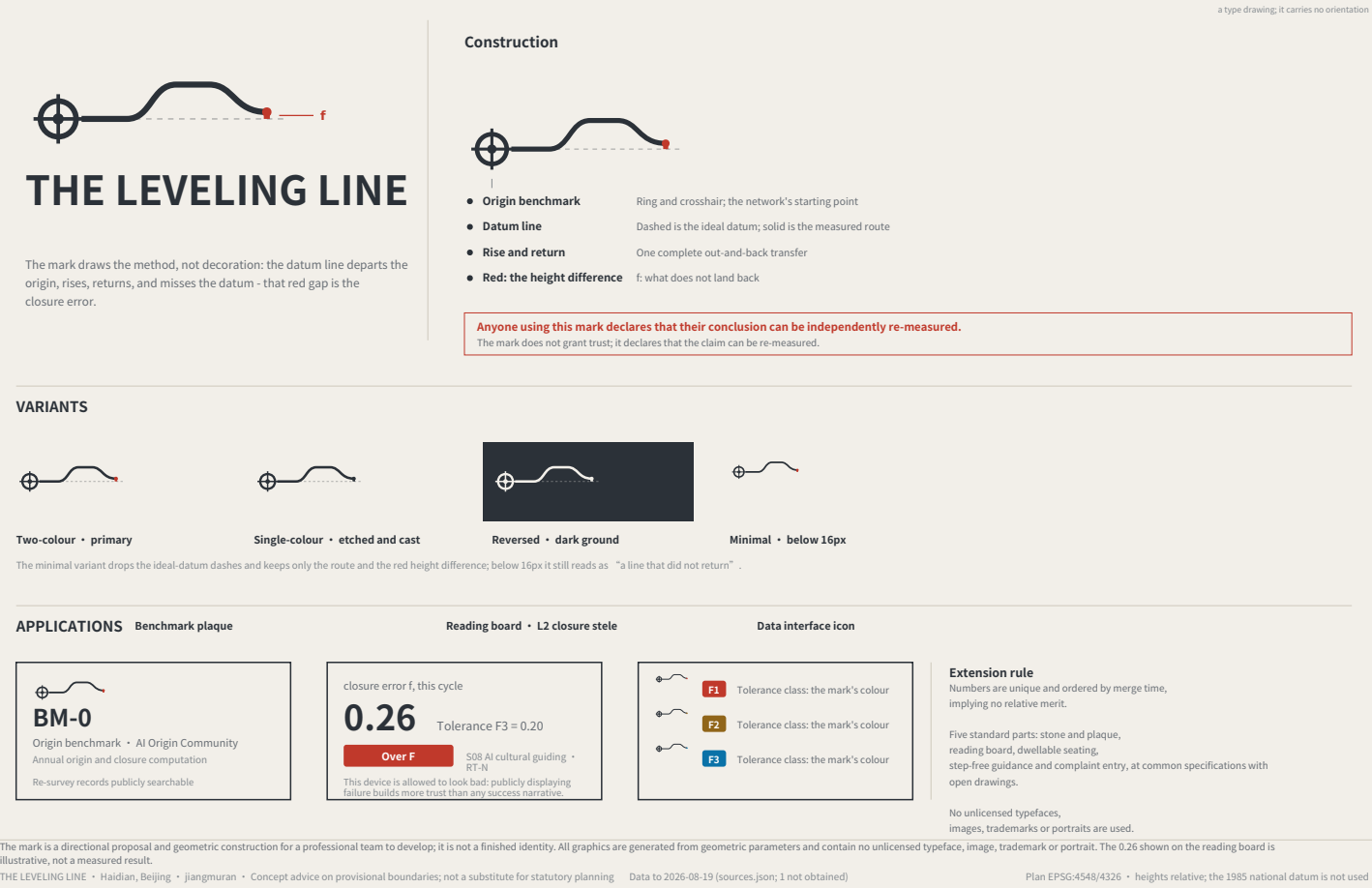
Run once: 10 cases, 2 accepted, 8 refused, reasons quoted from shipped files; 12 cards name the path that remains with the AI off.

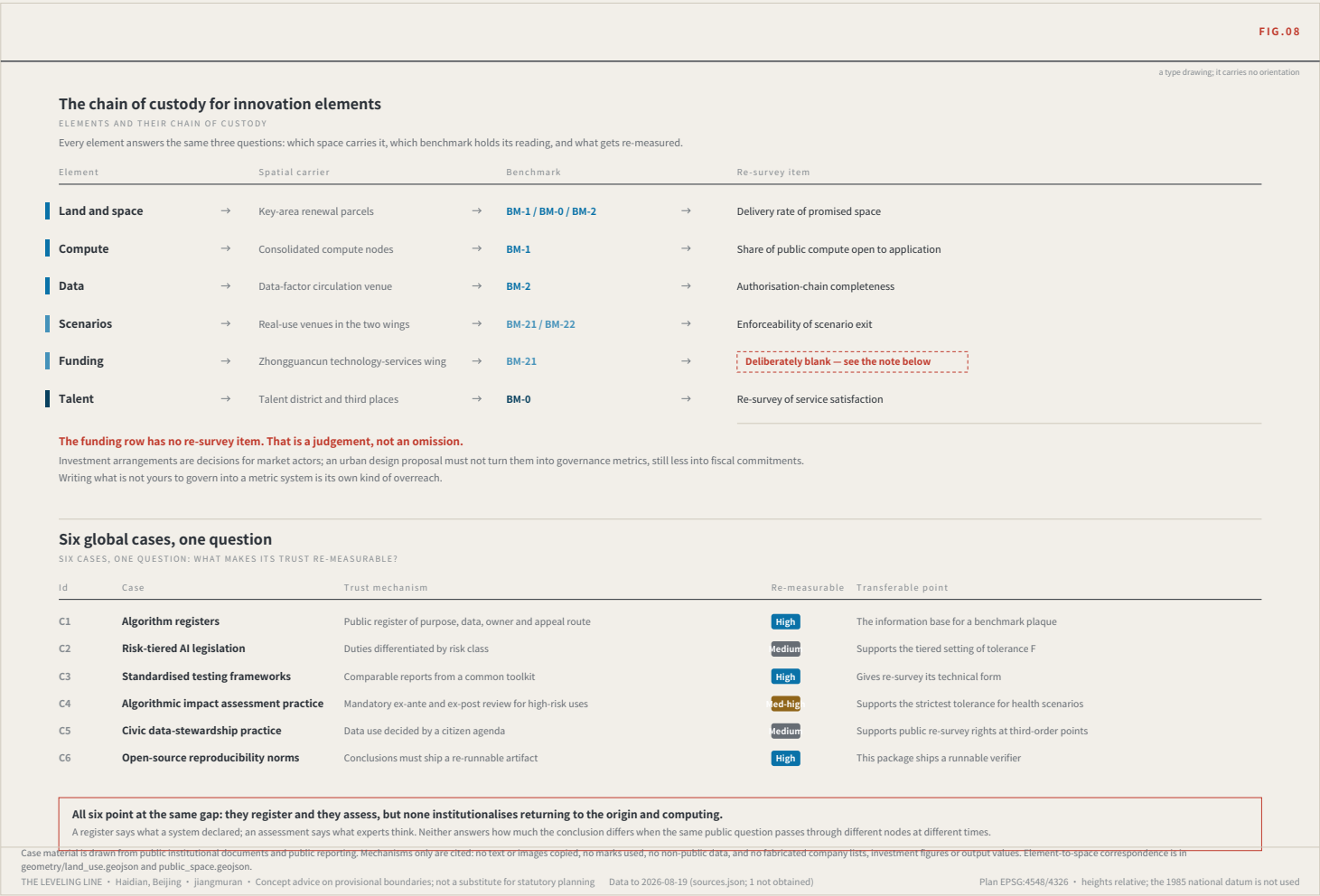
Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

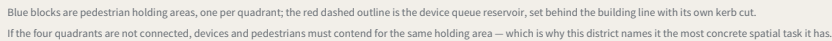
FIG.07





a type drawing; it carries no orientation

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty; carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources; jon; not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

DIMENSIONS: DESIGN INTENT vs AS-MEASURED

The right-hand column is deliberately empty.

Capacity is computed on site from measured clear width. A number filled in without measuring is manufactured certainty. Completed before any device is admitted.

KARB ALLOCATION SECTION



a type drawing; it carries no orientation



The steepest stretch is inside the planting, and 1:20 is the constraint on that stretch — an average gradient would hide it.

line.137 section says this edge needs a public route across the R&D block on the levee path, and no drawing showed it. This is that route: 6.0 m clear, centres at most 250 m apart (939.0 of them across a 4 m frontage), open 24 hours and ungated. The fallback alignment and the cost of a refusal are drawn. A type drawing, not a survey; dimensions are design intent, the as-measured column is as-surveyed; boundaries are provisional.

THE LEVELING LINE = Haidian, Beijing - jiangmuran - Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources,json; not obtained)

Plan EPSG:4548/4326 - heights relative; the 1985 national datum is not used

The right column is deliberately empty — effective clear width is filled in on site

The count is not a preference. It is a division.

Under the provisional substitute boundary the northern frontage of Zhongzhiyuan runs 939.0 m. At a maximum centre spacing of 250 m that is 4 public routes, 234.8 m apart. Replace the boundary with an official polygon and the number moves with it – it is computed from the geometry at build time, not written onto the sheet.

With the route open it is 133.5 m from the city road to the levee path. With route 2 refused and the walk diverted along the city road to route 1, it is 368.3 m. For P0.6, walking at 5 m/s, 3.7 minutes becomes 10.2 minutes. That is the working unit of a public route that can be locked.

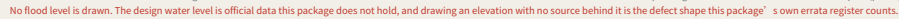
a type drawing; it carries no orientation

One horizontal scale • 220 m across each • no vertical exaggeration • ± 0.00 is that area's own benchmark, not a shared absolute datum

±0.00 = BM-1

192.9 ha • BM-1
dominant use R&D 100%
2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.



$\pm 0.00 = \text{BM-C}$

104.3 ha • BM-0
dominant use community services 94%
2 land-use classes inside

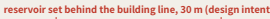
A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.



±0.00 = BM-2

72.0 ha • BM-2
dominant use industry and commerce 9
2 land-use classes inside

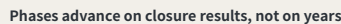
A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.



On the horizontal scale, one datum convention, no vertical exaggeration. They're placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is a step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line. ± 0.00 is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.

THE LEVELING LINE - Haidian, Beijing - jiangmuran - Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources: jom; not obtained) Plan EPSG:4548/4326 - heights relative; the 1985 national datum is not used

➤ north is right (unrolled)



The table above phases the eight points that exist and their cumulative cost. FIG.21 shows that meeting the fifteen-minute rule needs 9 more third-order points (a further CNY 22,005–76,358 a year), falling by order into the mid and long phases – this is not the phasing of a network that already complies.

The near-term increment is 32.1 ha and holds BM-0 alone: 1 re-survey a year, CNY 2,484–7,922 to run. That line is the answer to “this mechanism is too heavy” — it can start from one point, and that point costs what a sub-district office can approve by itself.

The mid phase does not begin because a second year arrived. It begins because the four near-term items are complete and $f \leq F$ for two consecutive cycles; the long phase the same. Once may be luck – the same rule this proposal writes at PATH-4 of the conversion path, applied to phasing. Fail it and the programme stays where it is, which makes the phasing table an exit table as well.

The condition that opens the mid phase reads “the four near-term items complete” and nothing in the package says which four — and the proposal contains two different sets of four: the near-term projects R1–R4, and the ones needing no official data, R1–R3 and R8. The second set contains a long-term item, so reading the gate against it makes the gate circular. Taking the first:

R1 datum stone L1 and the public evidence hall — one complete closure and its reading published within the first cycle

R2 first public setting of the tolerance F — a first version of $F1/F2/F3$ each, posted

R3 the four-week closure trial on cultural guidance S08 — closure recomputed and the consistency ratio published within four weeks

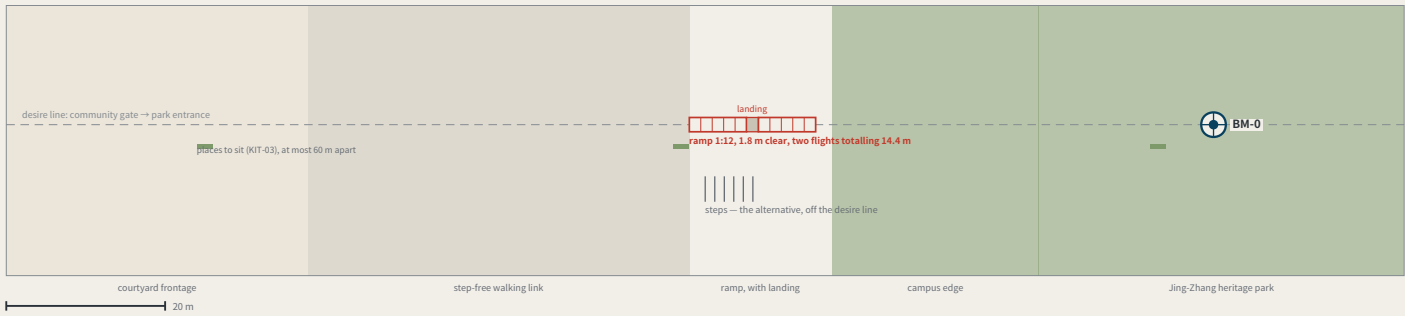
R4 third-order benchmarks set out at communities and rail stations — one recorded reading at each of the three community points

Dropping each benchmark into the phase polygons contradicts two of the phase names. The mid phase was named for “three first-order points” and contains two (BM-1, BM-2); the third is the datum BM-0, which is in the near phase. The long phase was named for “the full three-order network”, and the three third-order community points already fall inside the mid phase – what the long phase actually adds is the two second-order wing access points. The names were written for the intent and never checked against the polygons. Both are corrected in the layer, and every cell of this table is computed from the geometry rather than typed.

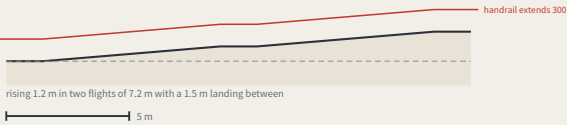
The phasing geojiao has shipped for a long time, been repaired once and enters recompute path, and no drawing had ever shown it. This sheet draws the three increments and costs each phase's year with the same model as build_operations, not a second set of coefficients. Phases open on closure results, not on orders. The geometry is concept advice on provisional boundaries and must be recomputed when official ones are published.

THE LEVELING LINE = Haidian, Beijing * jiangmunan * Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained) Plan EPSG:4548/4326 * heights relative; the 1985 national datum is not used

I. Plan



II. The ramp along its length (no vertical exaggeration)



Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why this number	As measured
Ramp gradient	1:12	step-free throughout; the ramp is on the desire line and the steps are its alternative	
Ramp clear width	≥ 1.8 m	two people passing, or a wheelchair with someone alongside	
landing	one every 750 mm of rise, at least 1.5 m long	a 1.2 m rise split into two flights — there has to be somewhere to stop	
Handrail	both sides, extending at least 300 mm past each end	a handrail that stops at the top of the slope stops where it is still needed	
Level approach at each end	at least 1.5 m level	level ground before and after, so nobody has to turn on the slope	
Spacing of places to sit	≤ 60 m	standard component KIT-03, 450 mm high with armrests	
Steps	beside the ramp, off the desire line	anyone who prefers steps is not sent around — steps are shorter than the ramp	

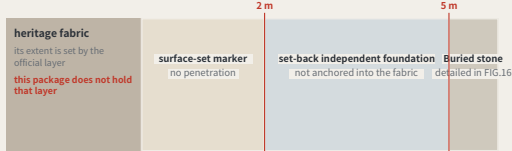
FIG.17's section says this place exists to put a 1:12 ramp where the steps are, and no drawing in the package had shown it. One design decision: the ramp sits on the desire line and the steps beside it — offset the ramp by 22 m and its user walks 44 m further every trip, while nobody on the steps walks a metre more. A type drawing, not a survey; dimensions are design intent and the as-built column is empty.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Monuments beside heritage fabric: the no-drilling construction and its setback rule

I. Choosing the construction by distance from the fabric



d = horizontal distance to the outer edge of the fabric

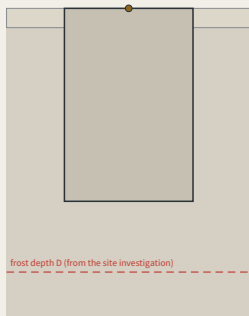
2 m

The rule is complete; its input is missing

Which line d is measured from depends on the officially published heritage protection area and its construction-control zone. This package does not hold that layer: each of the 3 shipped constraint features records it as a data gap in its own note, and this proposal does not infer it. So the sheet gives the rule, the thresholds and the three constructions, and leaves that line to whoever publishes it — guessing a protection boundary to make the drawing look complete would mean inventing the one thing here that is not this proposal's to decide.

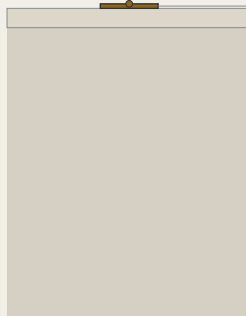
II. Two constructions, one scale

Buried: foundation below the frost line



can carry any order; removal needs plant

Surface-set: bonded, no drilling, no excavation



moves with the paving through the year → third order only, tied monthly

III. What each construction costs

Construction	Applies when	Foundation	Orders it may carry	Re-survey cycle	What removal leaves
Buried stone	$d \geq 5$ m	below the standard frost depth $D + 100$ mm	datum / first / second / third	per order	plant to lift out; leaves an excavation
set-back independent foundation	$2 \leq d < 5$ m	below frost line, set back ≥ 2 m, unanchored	second / third	per order	plant to lift out; leaves an excavation
Surface-set marker (no penetration)	$d < 2$ m	no foundation; bonded to existing paving, no drilling	third order only	monthly, and tied to the nearest buried point each time	lifting it leaves only a bond mark, which cleans off

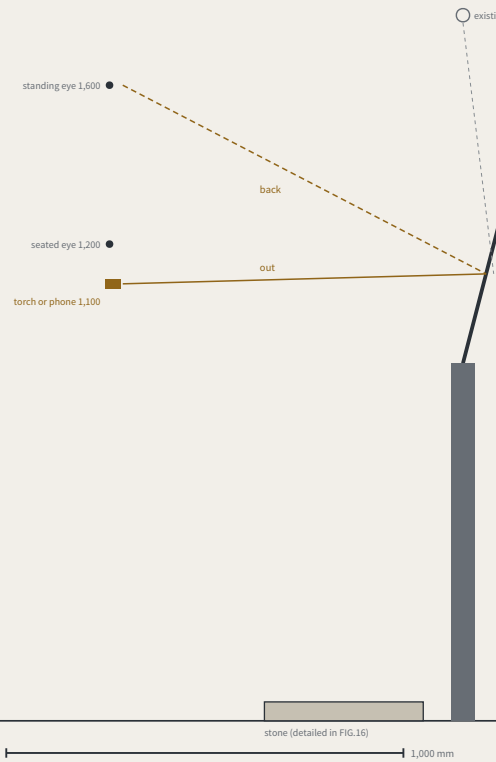
The call is for a belt along a hundred-year-old railway; this proposal's central act is sinking concrete. Three constructions, chosen by distance from the fabric, with the cost stated: without excavation there is no foundation below the frost line, so the surface-set marker may not carry first- or second-order duty. Where d is measured from depends on the official heritage layer, absent here and recorded as a gap in constraints.geojson.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Reading after dark, without lighting the benchmark

I. Geometry: where the light comes from, and returns to



II. Three approaches, compared by how they fail

Approach	Needs power	How it fails	Light trespass
Retroreflective face, reader brings the light	no	the face gets dirty → visible in daylight → caught before it stops being readable	none
Low-level floodlight	yes, at every point	the lamp dies → the plate looks intact and cannot be read	yes, needs controlling
Internally lit face	yes, at every point	the lamp dies → the whole face goes, and it still looks like a plate	yes, and aimed at the reader

The first is chosen not because it is cheap but because its failure is visible. When a lamp dies the plate still looks intact and can no longer be read — exactly the thing this proposal exists to prevent, a measurement whose failure cannot be seen. A dirty retroreflective face is simply dirty, visible in daylight to anyone walking past, well before it stops being readable.

plate face, raked 15° retroreflective film, returns light to its source

a specular surface would bounce it straight into the reader — hence not specular

III. What is not lit

No fixed luminaire	the reader brings the light; no power supply, no luminaire to maintain
The stone is not lit	it sits flush with the ground; lighting it adds no legibility, only night-time glare
Nothing aimed upward	no supplementary light may be emitted above the horizontal
Nothing crossing the boundary	the added vertical illuminance at neighbouring dwelling windows must meet the local limit — that figure comes from the local standard, which this package does not hold

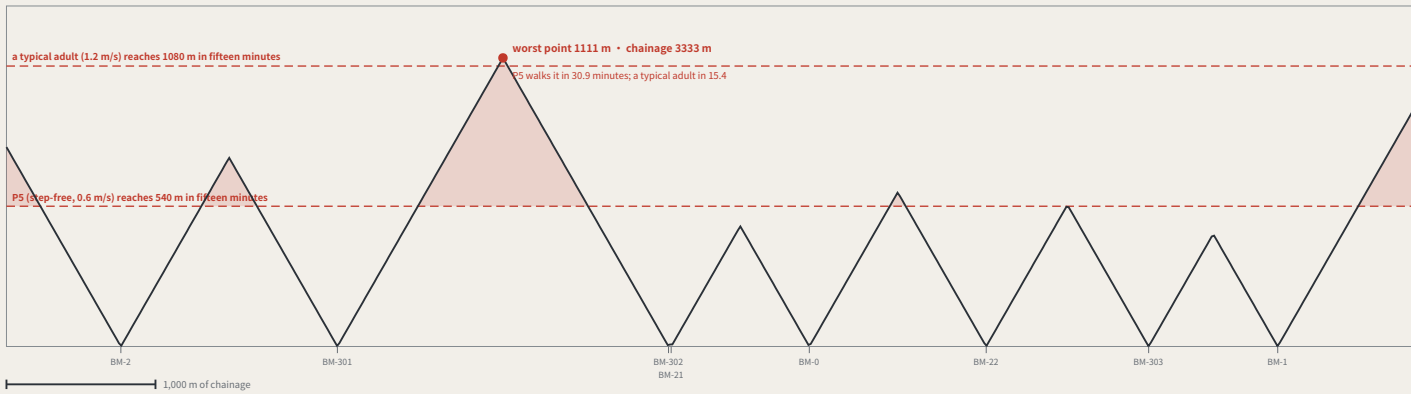
Three third-order points are read monthly by residents, and much of the year those days fall after dark — and the package had never said how the plate is read at night. The obvious answer is a lamp, and it is the wrong one: it needs power, needs maintaining, throws light into the windows opposite, and when it dies the plate looks intact and cannot be read — a measurement whose failure is invisible, which is the thing this proposal exists to prevent. So the face is retroreflective with no fixed luminaire and the reader brings the light; a dirty face is simply dirty, in daylight.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

How far the nearest benchmark actually is: this proposal's own rule, applied to itself

I. Distance to the nearest benchmark, along the spine



II. Turning that sentence into a spacing rule that can be checked

Where the rule comes from	this proposal's own repeated claim: a right of review you must walk fifteen minutes to reach has not been given
Interior spacing	chainage between adjacent points $\leq 2 \times 540$ m = 1080 m
Both ends of the line	end of line to its nearest point ≤ 540 m
Which speed the rule uses	PS's 0.6 m/s, not a typical adult's 1.2 m/s — setting the rule by the faster figure sets it at a standard most people cannot use

III. How far short the current layout falls

Six of the nine segments between the eight existing points fail. Closing the gap needs 9 more third-order points (or 3 if the rule is only held to an average walker). They are not free: third-order points are read monthly, and every one of them enters the annual operating table. This sheet does not place them — putting 9 coordinates on a provisional substitute spine is the manufactured certainty this register counts. The rule ships; the positions wait for the official alignment.

This proposal argues repeatedly that a right of review you must walk fifteen minutes to reach has not been given, and its entire participation argument rests on anyone being able to walk up to a benchmark and take a reading — and nobody had ever measured that walk. Measured: the worst place on the line is 1111 m from the nearest benchmark, 5 minutes for PS0.9, twice the limit this proposal holds others to. The rule ships as a spacing requirement; the positions wait for the official alignment.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

